EARTH OBSERVATION CROP RECOMMENDATION & DIVERSIFICATION

A transparent decision-support prototype for rain-fed and mixed crop–livestock farms in the Ben Ahmed study area, Morocco

PRECISION AGRICULTURE

GEOAI / EO

Challenge 813 • Space Academy Hackathon • Proof of Concept

Amine Fatah



The decision gap

Farm choices must reconcile water scarcity, crop suitability, livestock feed and regional resilience.

WATER

Rain-fed farms cannot treat every crop as equally feasible.

FEED

Mixed farms need crop plans that reduce livestock feed deficits.

ROTATION

Repeated crop families can weaken agronomic resilience.

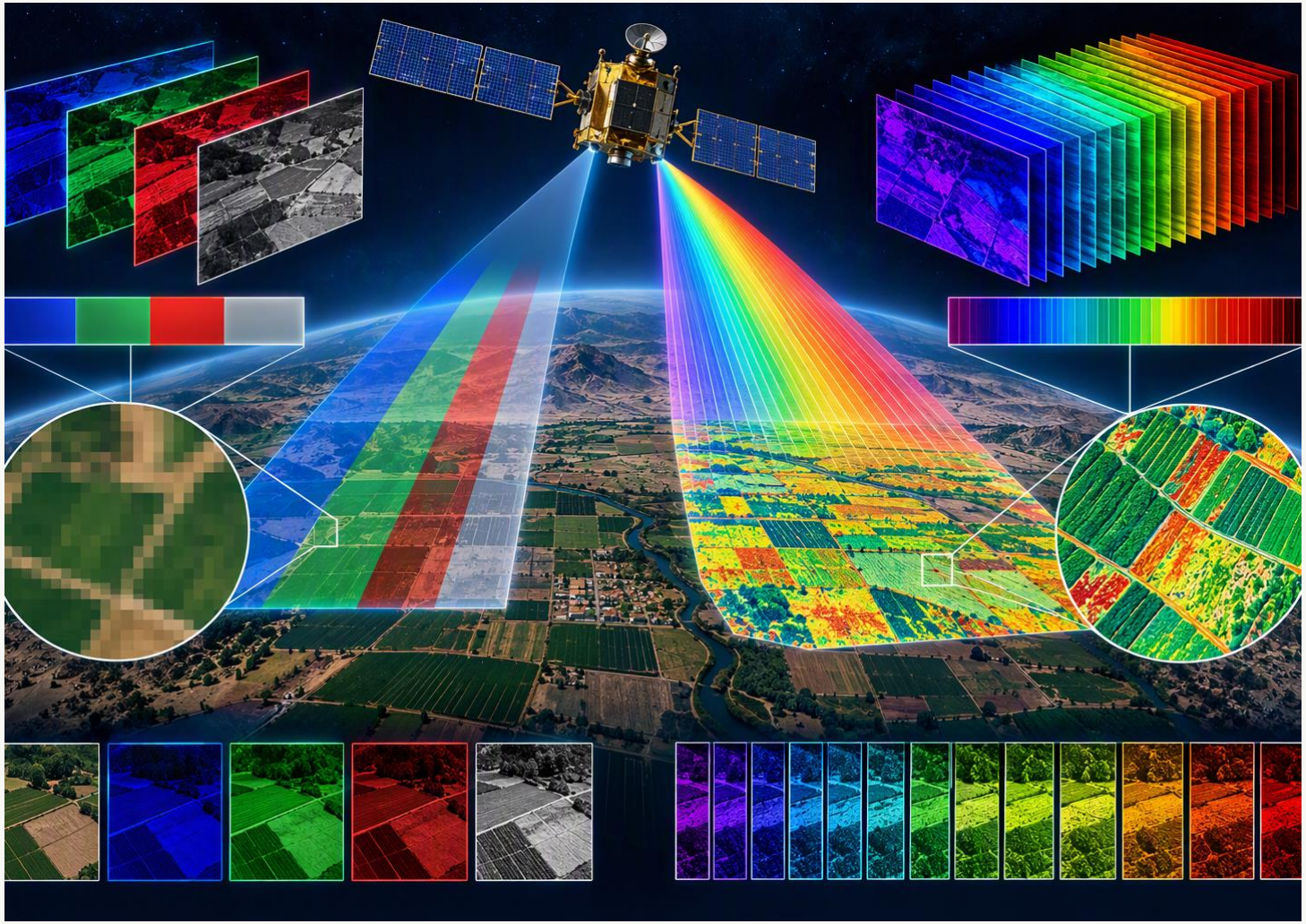
DIVERSITY

Individually rational choices can concentrate regional production.

Design principle **Make every recommendation explainable: show why a crop is eligible, why it scores well, and how much land it receives.**

Solution: from satellite pixels to farm decisions

A modular pipeline combines Earth observation, agricultural evidence and farmer constraints.



1 Observe

Sentinel-2 vegetation and moisture indicators

2 Contextualize

Climate, soil and GAEZ suitability

3 Personalize

Farm area, irrigation, objectives and livestock

4 Recommend

Eligible crop ranking and land allocation

5 Diversify

Regional share-aware scoring

Evidence stack

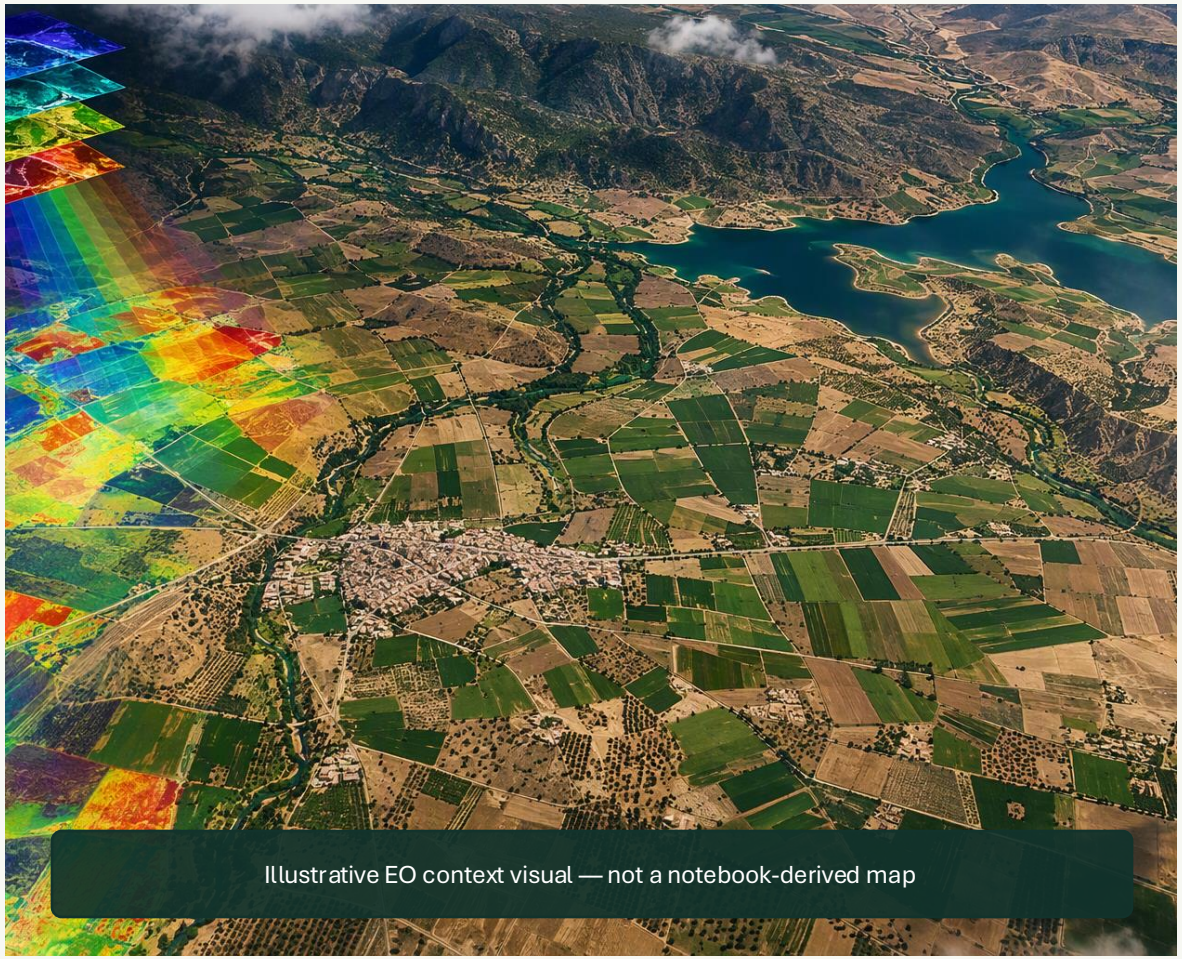
Each source answers a different part of the decision.

Sentinel-2 L2A	**Multispectral baseline**	True color • NDVI • NDMI	**10–20 m**
NASA POWER	**Regional climate**	Rainfall • temperature • solar	**Modelled**
SoilGrids	**Soil context**	Clay • sand • silt • organic carbon	**~250 m**
FAO–IIASA GAEZ v5	**Crop suitability**	Historical • rain-fed • low input	**Regional grid**
Farmer scenarios	**Decision constraints**	Area • irrigation • livestock • objectives	**12 synthetic farms**

Important: alignment creates a common analytical grid, not new parcel-level detail from coarse sources.

Earth-observation evidence in the current PoC

What is already implemented, what it supports, and what is deliberately not claimed.



CURRENT

Sentinel-2 Level-2A

True colour + NDVI + NDMI

Two-date multispectral baseline

CONTEXT

NASA POWER + SoilGrids

Climate and modelled soil

Regional evidence, not parcel truth

SCREEN

FAO-IIASA GAEZ v5

Rain-fed, low-input suitability

Scenario-specific crop evidence

NEXT

Challenge 813 hyperspectral

Narrow-band feature testing

Compare added value vs Sentinel-2

Scientific boundary: the current notebook demonstrates a multispectral decision-support baseline. It does not yet demonstrate hyperspectral added value or parcel-level farm prescriptions.

Transparent scoring and allocation

Eligibility first, then weighted scoring, then constrained allocation.



Outputs ranked crops • land shares • feed-gap context • rotation rationale • regional diversity metrics

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Prototype coverage

The notebook executes the complete decision flow across a controlled synthetic test set.

12

synthetic farmer scenarios

Rain-fed, irrigated, livestock and objective variations

8

candidate crops

Cereals, legumes and forage options

96

farmer–crop evaluations

Eligibility and scoring combinations

121.5 ha

allocated area

Complete allocation under both approaches

What the prototype demonstrates

- Every farmer keeps at least three eligible crop options.
- Both allocation approaches assign exactly 100% of farm area.
- Recommendations respond to irrigation, previous crop, objectives and feed need.
- Six crops receive land; chickpea and maize are excluded in the selected GAEZ scenario.

Case study: Farmer F03

A 10 ha rain-fed mixed farm with 20 sheep and a high estimated feed gap.



11.00 t
feed need

3.60 t
available

7.41 t
gap

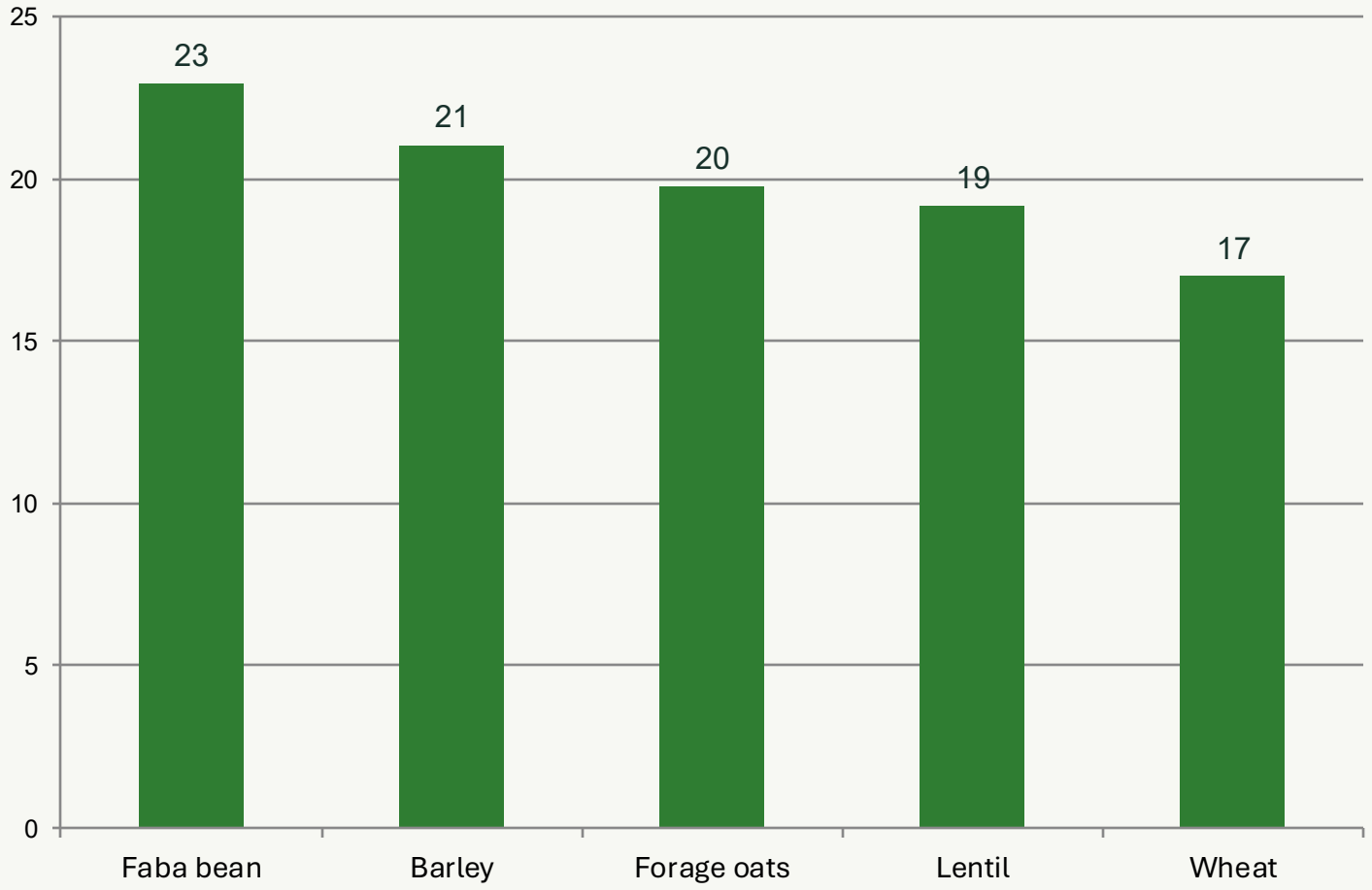
Recommendation logic

LEGUMES	Faba bean and lentil benefit from rotation after wheat.
FEED	Barley and forage oats respond to livestock demand.
WATER	Alfalfa and maize are excluded under no-irrigation conditions.
BALANCE	Five crops preserve food, feed and diversification functions.

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F03 recommendation card

Diversity-aware allocation keeps the full 10 ha farm productive across five crops.



TOP RANKED

Faba bean

Score ≈ 0.655

Why it wins

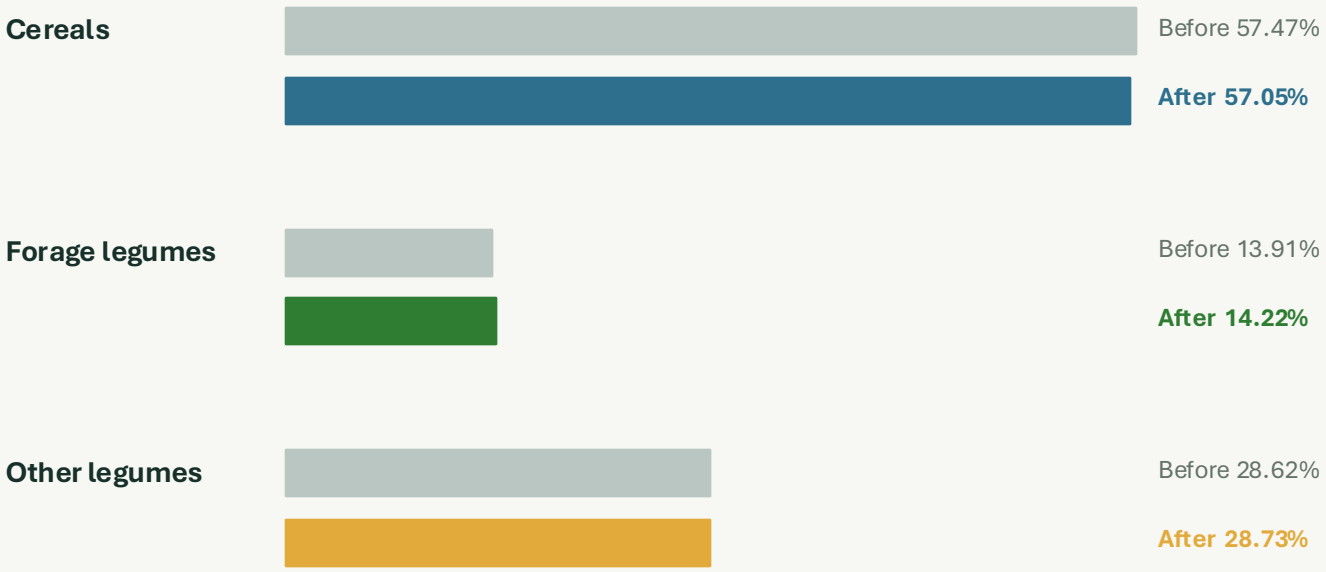
- Strong rotation benefit after wheat
- Food + moderate feed contribution
- Rain-fed compatible
- Adds a legume to the farm mix

100% allocated

Diversity-aware scoring: small but measurable effect

A secondary regional objective that never overrides environmental or farm constraints.

Regional crop-family share



Three complementary indicators

Shannon diversity	**1.7410 → 1.7426**	Higher is better
Effective crop count	**5.7033 → 5.7122**	Higher is better
HHI concentration	**0.1814 → 0.1809**	Lower is better

Interpretation The direction is positive but intentionally modest: diversity carries 10% of the score and cannot override suitability, water, livestock-feed or farmer-objective constraints. Faba bean remained the leading crop, so this is a nudge toward resilience, not forced equality.

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Challenge 813 fit and hyperspectral extension

A technically grounded PoC with a clear path from multispectral baseline to hyperspectral value.



OFFICIAL TRACK	Precision Agriculture & Smart Crop Analytics
EO-DRIVEN	Multi-source Earth observation translated into actionable decisions
SCALABLE	Modular data adapters and transparent farm-level scoring
813 UPSIDE	**Narrow-band stress and soil features can strengthen suitability evidence**
IMPACT	Supports food security, water efficiency and diversified production

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What is proven, and what is not

Credibility comes from keeping assumptions and validation needs visible.

DEMONSTRATED IN THE PoC

- Reproducible data loading and alignment
- Eligibility, scoring and full-area allocation
- Farmer-specific and diversity-aware outputs
- Traceable assumptions and explanations

CURRENT LIMITS

- Synthetic farmers and analytical boundary
- Four provisional crop-suitability values
- Regional climate/soil data, not parcel truth
- Feed coefficients are screening assumptions

VALIDATION NEEDED

- Local parcel boundaries and field observations
- Soil sampling and agronomic review
- Seasonal EO time series and yield evidence
- Farmer, livestock and market calibration

Positioning: decision-support prototype, not a validated farm prescription.

Builder and capability

A technically broad solo prototype with clear needs for domain validation and incubation support.

AMINE FATAH

Project builder

Earth observation • data engineering • explainable decision support

Ben Ahmed proof of concept

Technical	Python • data processing • geospatial workflow
Engineering	Mechanical engineering background • CATIA • MATLAB
Product mindset	Transparent recommendations • reproducible notebook
Needs to scale	Agronomist • livestock specialist • field partners

Collaboration ask: access to suitable hyperspectral data, agronomic validation, representative field observations, and incubation support.

Roadmap to an incubatable MVP

Convert the current notebook into a field-testable, hyperspectral-enhanced service.

NOW

PoC baseline

Notebook pipeline
12 scenarios
Explainable allocation

NEXT

813 integration

Acquire EnMAP / 813-compatible scene
Extract narrow-band features
Compare with Sentinel-2

VALIDATE

Field evidence

Parcels + soil samples
Agronomist review
Farmer feedback

MVP

Deployable service

GIQ-ready workflow
Scenario dashboard
Regional scaling

ASK: access to hyperspectral data, expert agronomic validation, and incubation support to turn the PoC into a scalable MVP.

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